MATHEMATICS

ON A THEOREM OF RABIN

BY

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In his recent paper [3], RABIN has shown that *the first-order theory of the system of all number theoretic predicates is categorical in cardinality* $\aleph_0$. Two general facts about certain denumerable relational systems will be given in this note each of which imply Rabin's result.

The relational systems considered here will all be of the form

$$\mathfrak{N} = \langle N, \oplus, \odot, 0, 1, R_0, \ldots, R_\xi, \ldots \rangle_{\xi<\alpha},$$

where $N = \{0, 1, 2, \ldots\}$ is the set of natural numbers; $\oplus$ and $\odot$ are two binary operations defined over N; 0 and 1 are the integers zero and one; and the R_ξ are additional (finitary) predicates over the domain N. The system $\mathfrak{N}$ will be called *standard* (for arithmetic) if $\oplus$ and $\odot$ are the ordinary operations $+$ and $\cdot$ of addition and multiplication over N. The system $\mathfrak{N}$ is *inductive* (or a model for arithmetic) if it satisfies the usual axioms for first-order arithmetic including all instances of the induction schema in which the symbols for the predicates R_ξ may occur in any combination desired. An inductive system is *non-archimedean* if it is not isomorphic to any standard system. The notions of *elementary equivalence* and *definability* as applied to relational systems are assumed known (see e.g. [3]). Finally, for each system $\mathfrak{N}$ introduce by recursion a function δ such that

(i) $\delta(0) = 0$;

(ii) $\delta(x+1) = \delta(x) \oplus 1$.

If $\mathfrak{N}$ is inductive, it is clear that we also have

(iii) $\delta(1) = 1$;

(iv) $\delta(x+y) = \delta(x) \oplus \delta(y)$;

(v) $\delta(x \cdot y) = \delta(x) \odot \delta(y)$.

Thus corresponding to $\mathfrak{N}$ there is a standard system $\mathfrak{N}^*$ determined by the condition that δ is an isomorphism from $\mathfrak{N}^*$ onto a subsystem of $\mathfrak{N}$. The additional relations of $\mathfrak{N}^*$ will be denoted by $R_\xi{}^*$, and hence

(vi) $R_\xi{}^*(x_0, \ldots, x_{n-1})$ *if and only if* $R_\xi(\delta(x_0), \ldots, \delta(x_{n-1}))$, in case the rank of R_ξ is n. There is, of course, no reason to believe that function δ and the relations $R_\xi{}^*$ are elementarily definable over $\mathfrak{R}$. With this notation and terminology we may now state the two theorems to be proved.

THEOREM 1. *If the two systems $\mathfrak{R}$ and $\mathfrak{S}$ are elementarily equivalent but not isomorphic, and if $\mathfrak{S}$ is a standard model for arithmetic, then the $\oplus$ and $\odot$ of $\mathfrak{R}$ are not elementarily definable over $\mathfrak{S}$.*

THEOREM 2. *If the system $\mathfrak{R}$ is inductive but non-archimedean, then the number of distinct predicates in the sequence $\langle R_0{}^*, \ldots, R_\xi{}^*, \ldots \rangle_{\xi<\alpha}$ is countable.*

Rabin's result is an immediate corollary of Theorem 1, for if $\mathfrak{S}$ is the system of all predicates over N, then every pair of operations is definable over $\mathfrak{S}$. But we may also apply Theorem 2. For let $\mathfrak{R}$ be equivalent to the standard $\mathfrak{S}$ with all predicates, then not only is $\mathfrak{R}$ inductive, but $\mathfrak{R}^* = \mathfrak{S}$. Hence, the sequence of the $R_\xi{}^*$ contains uncountably many relations, and by Theorem 2, $\mathfrak{R}$ is archimedean. This last statement means simply that $\mathfrak{R}^* = \mathfrak{S}$ is isomorphic to $\mathfrak{R}$.

The second argument has proved a stronger result: namely, we do not need the full force of the assumption that $\mathfrak{S}$ has all predicates. To be specific we may conclude that *the first-order theory of any uncountable system of number theoretic predicates uhich includes addition and multiplication is categorical in cardinality* $\aleph_0$. That this conclusion is actually stronger than Rabin's theorem was shown by SHOENFIELD in [4] where he announces the construction of an uncountable family of (monadic) predicates no one of which is even hyperarithmetically definable in any finite set of the remaining predicates. Thus a standard system $\mathfrak{S}$ formed by adjoining to $+$ and $\cdot$ a proper subset of Shoenfield's independent predicates will never be adequate for defining in first-order terms all predicates, and so Rabin's hypothesis can never apply to such systems.

Theorem 1 is a generalization of a result of FEFERMAN stated in [1], which was proved in a simpler way by TENNENBAUM in [5]. The diagonal argument used below is even more direct, however, and should be compared to Rabin's use of a diagonal argument. We may rephrase the content of this theorem as follows: *There is no arithmetically definable non-archimedean model for all true sentences of arithmetic even when we allow the system of arithmetic to contain any number of predicates beyond addition and multiplication.*

In giving formulas of the first-order theories relevant to the systems $\mathfrak{R}$, the symbols $\mathbf{+}$, $\cdot$, $\mathbf{0}$, $\mathbf{1}$, $\mathbf{R}_0, \ldots, \mathbf{R}_\xi, \ldots$, $\mathbf{=}$, V, $\mathit{\exists}$, $\wedge$, $\vee$, $\neg$, $\rightarrow$, $\leftrightarrow$ are used with their obvious import. The formulas $\mathbf{x} \leqslant \mathbf{y}$ and $\mathbf{x} | \mathbf{y}$ may be considered as abrevations of the formulas $\exists \mathbf{z}[\mathbf{x}+\mathbf{z}=\mathbf{y}]$ and $\exists \mathbf{z}[\mathbf{x}\cdot\mathbf{z}=\mathbf{y}]$, respectively. The following lemma is at the basis of the proofs of both of the theorems; its importance in this type of argument was first pointed out to the author by Tennenbaum. The idea is the same as that of the Chinese Remainder Theorem whose use to show the remarkable power of expression (in first-order terms) of $+$ and $\cdot$ of course goes back to Gödel. The proof can be found in MOSTOWSKI [2] and is easily adapted to the present formulation.

LEMMA (*First Form*). *Every inductive system satisfies all instances of the schema*

$$\forall \mathbf{b} \exists \mathbf{a}_0 \exists \mathbf{a}_1 \forall \mathbf{x}[\mathbf{x} \leqslant \mathbf{b} \rightarrow [(\mathbf{a}_0 \cdot (\mathbf{x}+\mathbf{1})+\mathbf{1}) | \mathbf{a}_1 \leftrightarrow \mathbf{\Phi}(\mathbf{x})]],$$

where the variables $\mathbf{a}_0$, $\mathbf{a}_1$, *are not free in the formula* $\mathbf{\Phi}(\mathbf{x})$.

For some purposes the two variables a_0 and a_1 are better condensed to one variable. The well-known polynomial pairing functions can be used to construct a formula $\mathbf{E}(\mathbf{x}, \mathbf{a})$, which represents a Diophantine predicate of two variables, and for which the Lemma takes the form:

LEMMA (*Second Form*). *Every inductive system satisfies all instances of the schema*

$$\forall \mathbf{b} \exists \mathbf{a} \forall \mathbf{x}[\mathbf{E}(\mathbf{x}, \mathbf{a}) \leftrightarrow [\mathbf{x} \leqslant \mathbf{b} \wedge \mathbf{\Phi}(\mathbf{x})]],$$

where $\mathbf{a}$ *is not free in* $\mathbf{\Phi}(\mathbf{x})$.

It must be stressed that the formula $\mathbf{\Phi}(\mathbf{x})$ may contain any of the symbols $\mathbf{R}_\xi$. The formula $\mathbf{E}(\mathbf{x}, \mathbf{a})$, whose exact nature is not important, does not contain the extra symbols, however. Obviously the idea of the Lemma is that single elements of an inductive system can "contain" information about a whole set of elements of the system. These sets must be bounded, but otherwise they can be constructed rather freely in terms of the basic relations. The exact meaning of the word "contain" is given by (the interpretation of) the formula $\mathbf{E}(\mathbf{x}, \mathbf{a})$.

Proof of Theorem 2. Let the system $\mathfrak{R}$ be inductive and non-archimedean. In the simplest case let us suppose that all the extra relations R_ξ are monadic predicates. Now $\mathfrak{R}^*$ is not isomorphic to $\mathfrak{R}$, so let b be an element of N not in the range of the function δ. It is easy to conclude that $\delta(x) \leqslant^\circ b$ for all x, where $\leqslant^\circ$ has its obvious definition in terms of $\oplus$. For each $\xi < \alpha$ we apply the Lemma (First Form) in the case where $\mathbf{\Phi}(\mathbf{x})$ is replaced by $\mathbf{R}_\xi(\mathbf{x})$. Hence, for each $\xi < \alpha$, there exist $a_0, a_1 \in N$ such that

$$R_\xi^* = \{x \in N : (a_0 \odot (\delta(x) \oplus 1)) \oplus 1 \,|^\circ a_1\},$$

where $|^\circ$ is defined in the obvious way in terms of $\odot$. Since the number of distinct pairs (a_0, a_1) is denumerable, the number of distinct R_ξ^* must be countable.

The assumption that the predicates R_ξ are monadic may easily be eliminated. Using the pairing functions, n-adic predicates may be made to correspond to monadic predicates, and the above argument shows that for each n there are only countably many predicates. Hence, the total number is countable.

Proof of Theorem 1. Suppose that $\mathfrak{S}$ is a standard system, and that $\mathfrak{R}$ is elementarily equivalent to $\mathfrak{S}$ but is not isomorphic to $\mathfrak{S}$. As we noticed before $\mathfrak{R}^* = \mathfrak{S}$, and $\mathfrak{R}$ is inductive and non-archimedean. By way of contradiction let us suppose that the $\oplus$ and $\odot$ of $\mathfrak{R}$ are elementarily definable over $\mathfrak{S}$. It follows that the interpretation in $\mathfrak{R}$ of the formula $\mathbf{E}(\mathbf{x}, \mathbf{a})$ yields an elementarily definable binary relation over $\mathfrak{S}$. Also since the function δ is primitive recursive in the function $\oplus$, the function δ is also definable over $\mathfrak{S}$. Let E be the relation over $\mathfrak{R}$ defined by the formula $\mathbf{E}(\mathbf{x}, \mathbf{a})$. Consider the set

$$D^* = \{x \in N:\ \textit{not}\ E(\delta(x), x)\}.$$

This set is definable over $\mathfrak{S}$ by some formula, $\mathbf{\Psi}(\mathbf{x})$ say. Let D be the set defined over $\mathfrak{R}$ by this same formula. By hypothesis $\mathfrak{R}$ and $\mathfrak{S}$ are equivalent, and so

$$D^* = \{x \in N:\ \delta(x) \in D\}.$$

Now choose b so that $\delta(x) \leqslant^\circ b$ for all $x \in N$ and apply the Lemma (Second Form) to obtain an element a such that

$$\{\delta(x) \in D:\ x \in N\} = \{\delta(x):\ E(\delta(x), a)\}.$$

Combining the three equations we find that for all $x \in N$,

$$\textit{not}\ E(\delta(x), x)\ \textit{if and only if}\ E(\delta(x), a).$$

The substitution of a for x gives the contradiction and completes the proof.

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